Department of Electrical and Electronics Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. EEE
Course Code & Title: EE8551 Microprocessors and Microcontrollers
Name of the Faculty member: Ms. S. Sharmila Kumari, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit I / Interrupts
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO 3
- **Activity Chosen: One minute paper**
- **Justification:**
  - Explain various types of interrupts
  - The topic interrupts in 8085 has 2 types - Hardware and Software. Since each type has its own definition, limitation and types. After teaching the concept, I thought of conducting this activity for making the students to give the difference between the each type of interrupts which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 5 Minutes
After teaching the concept, give students one or two minutes to think about the topic withoutwriting anything.
Total Strength is **33**
Reporter: **Myself**
At the end the Class (Last 5 minutes)

- ✓ I asked the students to think about various types of interrupts in 8085 concept for 3 minutes.
- ✓ Then I told them to write as much as they can within a short period of time (1 minute)
- ✓ Finally, I collected the papers from each column (1 minute)

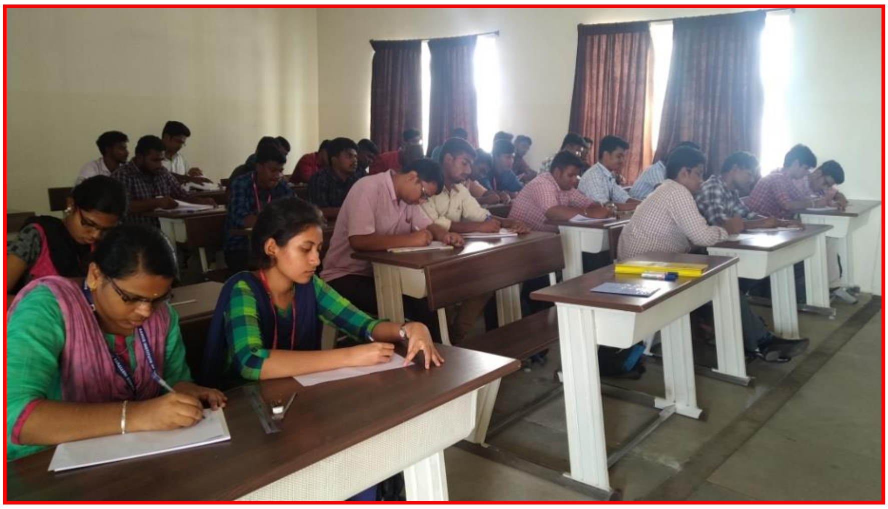
- **CO – PO / PSO mapping:**

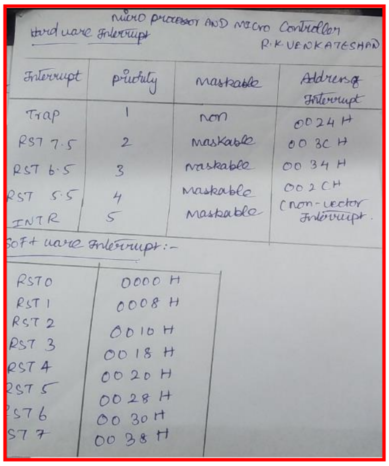
CO	PO1	PO2	PO3	PO10	PO12	PSO2	PSO3
C302.1	3	1	1	1	1	1	3

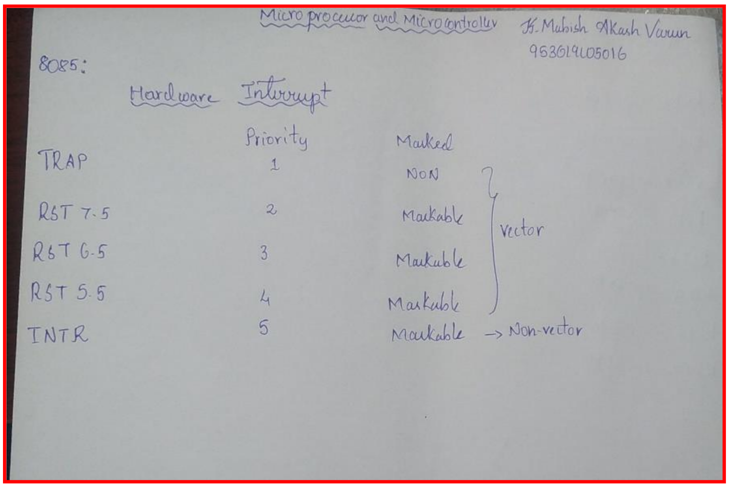
- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO10**	**PO12**	**PSO2**	**PSO3**
	3	**1**	**1**	**1**	**1**	**1**	**3**
Justification for correlation	Students will have fundamental electronic engineering concepts to solve electronic circuit problems.	Students will identify the mathematical, engineering and other relevant knowledge that applies to a memory organization of 8085 problem.	Students will use instruction cycle and timing diagram concept to understand the instruction process of 8085.	Students will be able to communicate the technical concept through Active Learning Methods.	Students will use the electronic fundamentals in the advancement of embedded system technology.	Students will be able to design and test the electronic system in the engineering applications.	Students will be able to design and develop the hardware and software with microprocessor skills required for industrial automation systems.

- **Images / Screenshot of the practice:**







❖ *Reflective Critique:*

1. ***Pre-implementation Reflection:***

- **Benefits:**
  - ➢ Students can able to attend the question even in the questions are in indirectform.
  - ➢ Students can able to explain the concepts in examination without any confusion.
- ***Challenges:***
  - ➢ In the class mostly boys hesitate to answer the questions.
  - ➢ Time utilization for conducting activity.

2. ***Post-implementation Reflection :***

- **Benefits:**
  - ➢ Students understood the concept which was reflected from their answers forthe questions I have asked during discussion session.
- ***Challenges:***
  - ➢ Slow learners were not able to understand some topics during discussionhours.

❖ *Benefit of the practice***:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

✓ The assessment of effectiveness of the activity was felt when told most of the points.

✓ While conducting the activity, I understood that the students will be able to explain various types of interrupts in 8085.

✓ The success of the activity was evaluated by asking the same question in Internal Assessment test I – Around 80% of students answered.

References:

❖ https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/184-visual-modeling-one-minute-paper

Signature of Faculty Member **HOD**

Department of Electrical and Electronics Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. EEE
Course Code & Title: EE8551 Microprocessors and Microcontrollers
Name of the Faculty member: Ms. S. Sharmila Kumari, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit II / Addressing Modes of 8085
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 4
- **Activity Chosen: Mini - map**
- **Justification:**
  - Differentiate the type of Addressing modes of 8085
  - After teaching the concept, I thought of conducting this activity for making the students to give the difference between the 5 types of addressing mode which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 6 Minutes

After teaching the concept, the students were made to pair with their neighbors

Photographer: Myself

Reporter: Myself

At the end the Class (Last 6 minutes)

- ✓ I asked the students to think about types of 8085 addressing modes concept for 2 minute.
- ✓ Then I told them to Pair with their neighbors and discuss about the concepts for another 1 minute.
- ✓ Finally I told them to design mini-map and submit within 3 minutes.

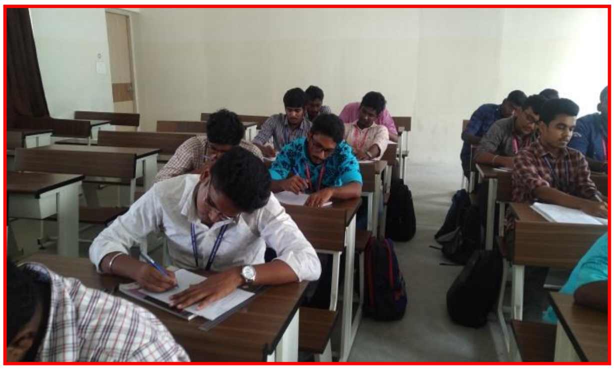
- **CO – PO / PSO mapping:**

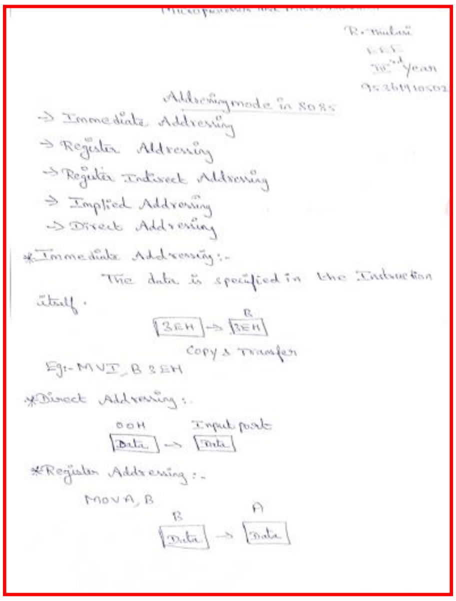
CO	PO1	PO2	PO3	PO10	PO12	PSO2	PSO3
C302.2	3	2	3	1	1	1	3

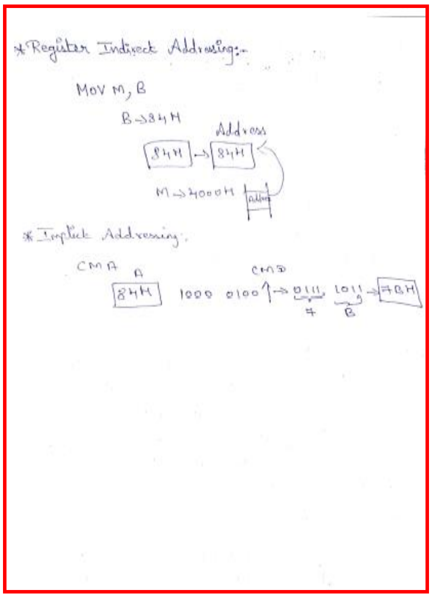
- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO10	PO12	PSO2	PSO3
	3	2	3	1	1	1	3
Justification for correlation	Students will have fundamental electronic engineering concepts to solve electronic circuit problems.	Students will identify the Instruction set and Assembly language format of 8085.	Students will determine the Instruction format and Addressing modes of 8085 microprocessor.	Students will be able to communicate the technical concept through Active Learning Methods.	Students will use the electronic fundamentals in the advancement of embedded system technology.	Students will be able to design and test the electronic system in the engineering applications.	Students will be able to design and develop the hardware and software with microprocessor skills required for industrial automation systems.

- **Images / Screenshot of the practice:**







❖ *Reflective Critique:*

1. ***Pre-implementation Reflection:***

I preferred this Activity because they have five different types of 8085 addressing modes separately.

Challenges anticipated:

- In the class mostly boys hastate to answer to the questions.
- Time utilization for conducting activity.

Steps taken:

- The boys are sitting in 2 columns – I planned to choose more pairs from boysto involve them in the activity.

2. ***Post implementation Reflection:***

Benefits:

- Students understood the concept which was reflected from their answers forthe questions I have asked during discussion session.

Challenges:

- Slow learners were not able to understand some topics during discussionhours.

❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

✓ The assessment of effectiveness of the activity was felt when told most of the points.

✓ While conducting the activity, I understood that the students will be able to explain the concepts of Addressing modes.

✓ The success of the activity was evaluated by asking the same question in Internal Assessment test I – Around 50% of students answered.

References:

1. https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/184-visual-modeling-mini-maps

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Innovative Practice Description

- **Unit / Topic:** Unit III / Compare the programming concept of 8051 with 8085
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 10
- **Activity Chosen: Visual Clicker**
- **Justification:**
  1. Differentiate the type of Addressing modes of 8085
     - After teaching the concept, I thought of conducting this activity for making the students to give the difference between the 5 types of addressing mode which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 6 Minutes

After teaching the concept of 8051 and 8085, the students were made to pairwith their neighbors
Total Strength is 33,
Number of Pairs – minimum 3 in one team
Photographer: one student – Mr. S. Jeyasurya (interested in photography)
Reporter: Myself
At the end the Class (Last 6 minutes)

- ✓ I asked the students to think about 8051 and 8085 concept for 1 minute.
- ✓ Then I told them to Pair with their neighbors and discuss for another 1 minute.
- ✓ With the help of PPT prepared (Multiple choice questions) students use their clickersto respond to the question and then the answers were be projected to the students.
- ✓ After completion of the class I uploaded the PPT in the course website (Canvas)

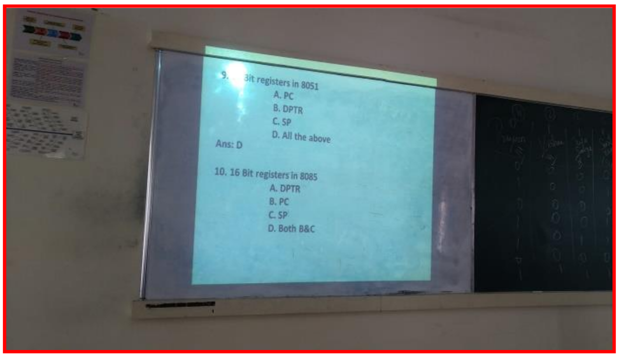
- **CO – PO / PSO mapping:**

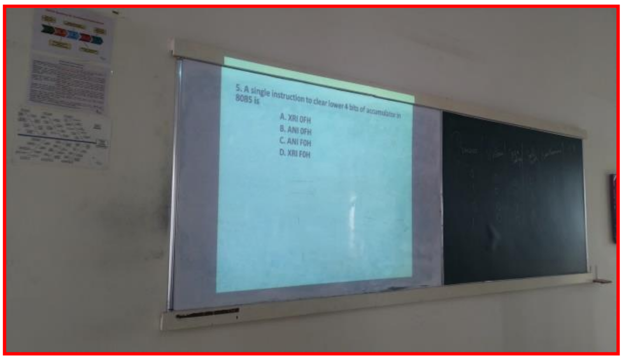
CO	PO1	PO2	PO3	PO9	PO10	PO12	PSO2	PSO3
C302.3	3	1	1	1	1	1	1	3

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO9**	**PO10**	**PO12**	**PSO2**	**PSO3**
	3	**1**	**1**	**1**	**1**	**1**	**1**	**3**
Justification for correlation	Students will have fundamental electronic engineering concepts to solve electronic circuit problems.	Students will identify the memory organization of 8051 problems, Instruction set and Assembly language format of 8051.	Students will use instruction cycle and timing diagram concept to understand the instruction process of 8051 and also able to compare the programming concept of 8051 microcontroller with 8085 microprocessor.	Students will be able to work with their team through Active Learning Methods.	Students will be able to communicate the technical concept through Active Learning Methods.	Students will use the electronic fundamentals in the advancement of embedded system technology.	Students will be able to design and test the electronic system in the engineering applications.	Students will be able to design and develop the hardware and software with microprocesso skills requirec for industrial automation systems.

- **Images / Screenshot of the practice:**





❖ *Reflective Critique:*

1. Pre-implementation Reflection:

- I preferred this Activity because they have 8051 and 8085separately.

Challenges anticipated:

- Time utilization for conducting activity.

2. Post implementation Reflection:

Benefits:

- Students understood the concept which was reflected from their answers forthe questions I have asked during discussion session.

Challenges:

- Student clicker's were not clear in photos. So clicker tools prepared fromstudents were not advisable.

❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

✓ The assessment of effectiveness of the activity was felt when told most of the points.

✓ While conducting the activity, I understood that the students will be able to explainthe concepts of 8085 and 8051.

✓ The success of the activity was evaluated by asking the same question in InternalAssessment test II – Around 50% of students answered.

References:

1. https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27- teaching/184-visual-modeling

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Innovative Practice Description

- **Unit / Topic:** Unit IV / A/D and D/A Converters & Interfacing with 8051.
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 15
- **Activity Chosen: Flipped Classroom**
- **Justification:**
    - Flipped classroom is a "pedagogical approach in which direct instruction moves from the group learning space to the individual learning space, and the resulting group space is transformed into a dynamic, interactive learning environment where the educator guides students as they apply concepts and engage creatively in the subject matter"

Time Allotted for the Activity: 30 Minutes

Number of Students – 4

Photographer: one student – Mr. S. Jeyasurya (interested in photography)

Reporter: Myself

At the Class (30 minutes)

- I shared the material related to the topic through canvas LMS.
- Then I told them to prepare the topic as a team with a help of PPT.

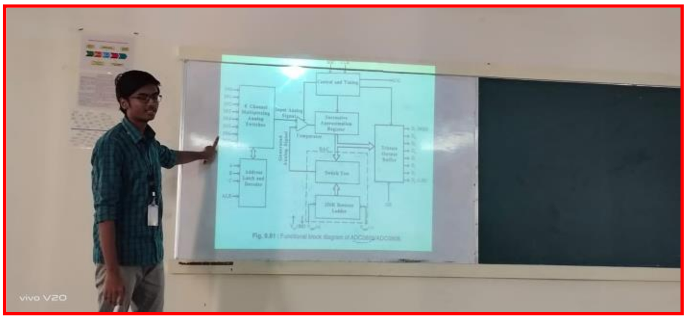
- **CO – PO / PSO mapping:**

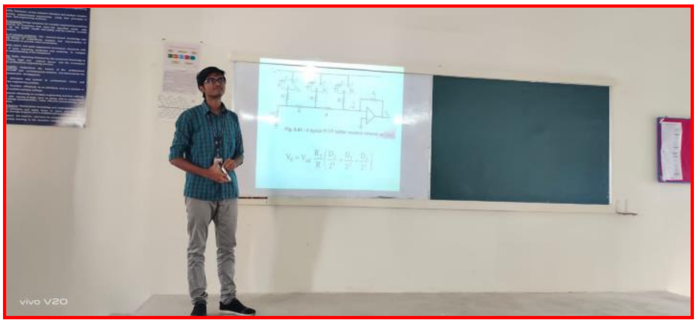
CO	PO1	PO2	PO3	PO9	PO10	PO12	PSO2	PSO3
C302.4	3	2	3	1	1	1	1	3

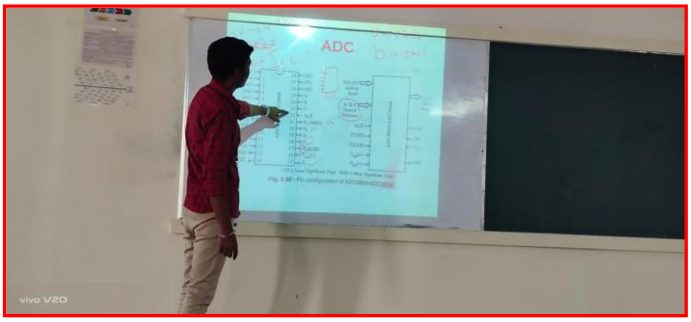
- **PO / PSO mapped:**

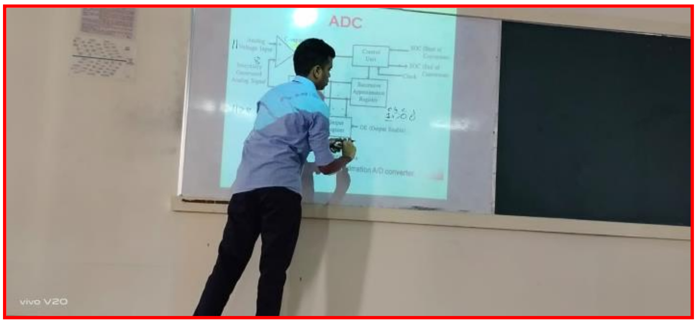
Innovative practice	**PO1**	**PO2**	**PO3**	**PO9**	**PO10**	**PO12**	**PSO2**	**PSO3**
	3	**1**	**1**	**1**	**1**	**1**	**1**	**3**
Justification for correlation	Students will have fundamental electronic engineering concepts to solve electronic circuit problems.	Students will identify the design specification of basic interfacing circuits with 8085 and 8051.	Students will determine the design objectives and functional requirements of the basic interfacing circuits with 8085 and 8051.	Students will be able to work with their team through Active Learning Methods.	Students will be able to communicate the technical concept through Active Learning Methods.	Students will use the electronic fundamentals in the advancement of embedded system technology.	Students will be able to design and test the electronic system in the engineering applications.	Students will be able to design and develop the hardware and software with microprocessor skills required for industrial automation systems.

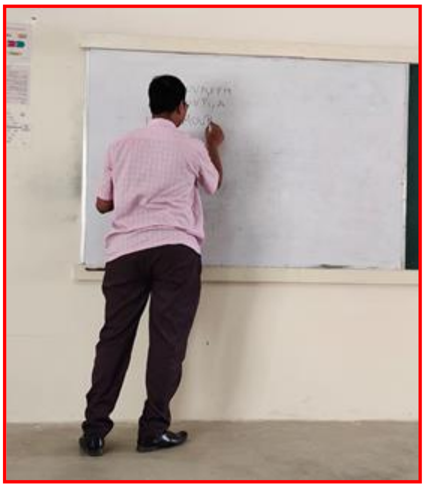
- **Images / Screenshot of the practice:**











❖ *Reflective Critique:*

Implementation Reflection:

- **Benefits:**
  - ➢ This practice improves the peer coaching / studying with in the class.
  - ➢ Students understood the concept which was reflected from their answers forthe questions I have asked during discussion session.

References:

1. https://www.panopto.com/blog/7-unique-flipped-classroom-models-right/

Signature of Faculty Member **HOD**